VM networking modes (NAT, NAT Network, Host-only, Bridged): when to use each, and why vulnerable labs must never be bridged.

Virtualization allows multiple virtual machines (VMs) to run on one physical computer using software such as Oracle VirtualBox. Each VM can run its own operating system, such as Kali Linux, Ubuntu, or Windows.

VM networking determines how virtual machines communicate with one another, the host computer, and external networks.

The four common VirtualBox networking modes are NAT, NAT Network, Host-only, and Bridged networking.

NAT (Network Address Translation)

Definition: NAT allows a virtual machine to access external networks through the host computer's network connection. The VM typically uses a private virtual address, and VirtualBox translates its traffic.

When to use it:

- Downloading updates for a VM.
- Installing software that requires internet access.
- Giving one VM outbound connectivity without making it directly accessible from the physical LAN.

Example: A Kali Linux VM uses NAT to download package updates while unsolicited inbound connections to the VM are generally restricted by default.

Security consideration: NAT reduces direct inbound exposure but does not guarantee complete isolation. A compromised VM may still communicate with external systems.

NAT Network

Definition: NAT Network allows multiple VMs to communicate with one another on a shared virtual network while also providing outbound network access through NAT, depending on the configuration.

When to use it:

- Building a multi-VM cybersecurity lab.
- Connecting Kali Linux and Ubuntu for controlled exercises.
- Allowing lab VMs to communicate while retaining outbound internet access.

Example: Kali Linux and Ubuntu share a NAT Network, allowing them to communicate with each other and download updates.

Security consideration: VMs on the same NAT Network can generally communicate with one another. Use appropriate firewall rules and avoid connecting vulnerable machines to unnecessary networks.

Host-only Networking

Definition: Host-only networking creates a private virtual network connecting the host computer and selected VMs. It normally does not provide direct internet access through that adapter.

When to use it:

- Practising networking in an isolated environment.
- Testing connectivity between your own VMs.
- Learning Linux administration and security without exposing lab services to the physical LAN.

Example: Kali Linux and Ubuntu use a Host-only network to exchange traffic while remaining separate from the household network.

Security consideration: Host-only networking improves isolation from the physical LAN, but the host and other attached VMs can still access the virtual network. It should not be treated as a guarantee against all forms of compromise.

Bridged Networking

Definition: Bridged networking connects a VM to the physical network through the host's network interface. The VM behaves more like a separate device on the same LAN and may receive an address from the physical network's DHCP server.

When to use it:

- Testing a VM as a normal network device in an authorized environment.
- Performing specific LAN compatibility tests.
- Practising network administration when direct LAN participation is necessary and approved.

Example: An Ubuntu VM receives an IP address from the home router and communicates with other devices on the home LAN.

Security consideration: The VM may be reachable by other devices on the physical network, depending on firewall rules and service configuration.

Feature	NAT	NAT Network	Host-only	Bridged
Internet access	Usually yes	Usually yes	Normally no, through that adapter alone	Usually yes, if the LAN provides access
VM-to-VM communication	Not generally available by default between separate NAT-attached VMs	Yes, on the same network	Yes, on the same host-only network	Yes, subject to network configuration
Host-to-VM communication	Restricted by default; port forwarding may be needed	Depends on configuration and host connectivity	Yes, normally	Usually, subject to LAN and firewall rules
Physical LAN exposure	Lower direct exposure	Lower direct exposure, but outbound communication remains possible	Normally isolated from the physical LAN	Direct participation in the physical LAN
Common use	Internet access for one VM	Multi-VM lab with outbound access	Isolated local lab	Authorized LAN testing

Why Must Vulnerable Lab VMs Never Be Bridged?

Intentionally vulnerable VMs are designed for security practice. They may contain outdated software, weak passwords, insecure services, or known vulnerabilities.

Bridging these machines to a real physical network can expose them to household devices, workplace systems, and other devices on the same LAN.

Main risks

- **Unauthorized access:** Other devices on the LAN may be able to connect to vulnerable services.
- **Malware spread:** A compromised VM could become a stepping stone toward other systems.
- **Data exposure:** Shared network access may expose files or services unintentionally.
- **Accidental testing:** Scans or security tests may reach devices outside the intended lab.
- **Network disruption:** Vulnerable services or poorly controlled testing could affect real devices.

Recommended safety practices

- Prefer Host-only networking for an isolated lab.
- Use NAT Network when multiple lab VMs need to communicate and outbound access is necessary.
- Keep vulnerable targets off Bridged networking.
- Enable host and guest firewalls where appropriate.
- Disable unnecessary shared folders, shared clipboard, and drag-and-drop for vulnerable VMs.
- Take clean snapshots before testing and revert when necessary.
- Verify adapter settings and network routes before starting practical exercises.

Conclusion

VirtualBox networking modes control how virtual machines communicate with the host, each other, and external networks.

NAT provides outbound connectivity, NAT Network supports communication between VMs on a shared NAT-backed network, Host-only provides a private lab network, and Bridged connects a VM to the physical LAN.

For cybersecurity practice, Host-only or a carefully configured NAT Network is generally preferable to Bridged networking because it reduces the chance of exposing vulnerable machines to real devices.